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POWER CONVERTERS WITH WIDE BANDGAP SEMICONDUCTORS

Abstract

A power supply system including a housing, an internal non-removable battery, an output, a power converter, and an electronic processor. The internal non-removable battery received in the housing and having a nominal voltage of at least 12 Volts. The output provided on the housing. The power converter coupled between the internal non-removable battery and the output. The power converter including a first switch. The power converter also converts power received from the internal non-removable battery. The electronic processor coupled to the power converter and controls operation of the power converter by controlling the first switch at a switching frequency of at least 100 kHz.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of and claims the benefit of U.S. patent application Ser. No. 18/594,775, filed Mar. 4, 2024, which claims priority to U.S. patent application Ser. No. 17/050,856, filed Oct. 27, 2020, which granted as U.S. Pat. No. 11,923,716 on Mar. 5, 2024, which claims priority to U.S. Provisional Patent Application No. 62/900,252, filed on Sep. 13, 2019, the entire content of which is hereby incorporated by reference.

FIELD

[0002] The present disclosure relates to power converter devices such as chargers, AC/DC adapters, battery powered mobile power supplies, and the like.

BACKGROUND

[0003] Several kinds of cordless direct-current (DC) power tools and corded alternating-current (AC) power tools are used at construction sites and other locations. Operating power may not always be available in both AC and DC forms at a particular location, and a user may have a power tool that uses a power form not available. For example, the user may have an AC power tool at a location without AC power available.

SUMMARY

[0004] Power converting devices can convert power available in one form (e.g., DC) to power in another form (e.g., AC) to enable power tools and other devices to operate regardless of the form of power available at a particular location. However, power converting devices include internal converters that are bulky, heavy, and lack efficiency during operation. For example, some portable power supplies use an inverter including several metal oxide semiconductor field effect transistors (MOSFETs) for converting power from DC to AC. MOSFET based designs are large, heavy, and lose high amounts of energy as heat when operated at high-frequencies. To get more power out of the MOSFET based designs, additional circuitry and heat sinking features are used that add to the size and weight of the portable power supply. Additionally, MOSFETs cannot operate efficiently at high switching frequencies. This, in turn, results in the use of large passive components due to low frequency operation of MOSFETs. Other power converter devices have similar problems when using MOSFETs in power conversion circuitry.

[0005] Accordingly, there is a need for power converting solutions that are smaller, lighter, and more efficient, and a need for switches in power converting devices that are more efficient at high-frequencies.

[0006] Portable power sources described herein include a housing, a battery coupled on the housing, an outlet provided on the housing and configured to provide power to a device external to the portable power source, a power converter coupled between the battery and the outlet and configured to convert power received from the battery. The power converter includes a first wide bandgap field effect transistor (FET). The portable power sources also include an electronic processor coupled to the power converter and configured control operation of the power converter by controlling the first wide bandgap FET.

[0007] In some aspects, the battery has a nominal voltage between 12 Volts (V) and 680 V. In some aspects, the nominal voltage is between 12 V and 1200 V.

[0008] In some aspects, the housing includes a battery pack interface to receive the battery, and

wherein the battery is a power tool battery pack having a nominal voltage selected from a group consisting of: 12 V, 18 V, 60 V, 80 V, and between 12 V and 80 V.

[0009] In some aspects, the power converter includes an inverter including the first wide bandgap FET and is configured to convert direct-current (DC) power received from the battery to alternating-current (AC) power, wherein the AC power is provided at the outlet.

[0010] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency of at least 100 kilo-Hertz (kHz). In some aspects, the switching frequency is at least 125 kHz.

[0011] In some aspects, the power converter further includes a DC-to-DC converter configured to convert DC power received from the battery pack at a first voltage to DC power at a second voltage, and provide DC power at the second voltage to the inverter, wherein the DC-to-DC converter includes a second wide bandgap FET.

[0012] In some aspects, the electronic processor controls the second wide bandgap FET at a switching frequency of at least 200 kHz. In some aspects, the switching frequency is at least 400 kHz.

[0013] In some aspects, the inverter includes a first bridge circuit having a first high-side FET and a first low-side FET, and a second bridge circuit having a second high-side FET and a second low-side FET, wherein at least one of the first high-side FET, the first low-side FET, the second high-side FET, and the second low-side FET is the first wide bandgap FET, and wherein at least another of the first high-side FET, the first low-side FET, the second high-side FET, and the second low-side FET is a metal oxide semiconductor field effect transistor (MOSFET).

[0014] In some aspects, the first high-side FET is the first wide bandgap FET and the first low-side FET is a second wide bandgap FET, and the second high-side FET and the second low-side FET are MOSFETs.

[0015] In some aspects, the first high-side FET is the first wide bandgap FET and the second high-side FET is a second wide bandgap FET, and wherein the first low-side FET and the second low-side FET are MOSFETs.

[0016] In some aspects, the first low-side FET is the first wide bandgap FET and the second low-side FET is a second wide bandgap FET, and wherein the first high-side FET and the second high-side FET are MOSFETs.

[0017] In some aspects, the electronic processor operates the first wide bandgap FET at a first frequency and operates the MOSFET at a second frequency, and wherein the first frequency is higher than a second frequency.

[0018] In some aspects, the power converter includes a DC-to-DC converter having the first wide bandgap FET and is configured to convert DC power received from the battery pack at a first voltage to a second voltage, and the DC power at the second voltage is provided at the outlet.

[0019] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency of at least 200 (kHz). In some aspects, the switching frequency is at least 400 kHz.

[0020] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency between 100 kHz and 400 kHz.

[0021] In some aspects, the portable power source includes a power input provided on the housing to receive AC power from an external source and a second power converter coupled between the power input and the battery and configured to convert power received from the power input. The power converter includes a second wide bandgap field effect transistor (FET).

[0022] In some aspects, the second power converter includes an AC-to-DC converter having the first wide bandgap FET and configured to convert AC power received from the power input to DC power provided to the battery for charging the battery.

[0023] Power tool battery pack chargers described herein provide a housing, a power input provided on the housing to receive AC power from an external source, a first battery pack interface provided on the housing and configured to receive a first-type of power tool battery pack, and a

power converter coupled between the power input and the first battery pack interface and configured to convert power received from the power input. The power converter includes a first wide bandgap field effect transistor (FET). The power tool battery pack chargers also include an electronic processor coupled to the power converter and configured control operation of the power converter by controlling the first wide bandgap FET.

[0024] In some aspects, a second battery pack interface is provided on the housing and is configured to receive a second-type of power tool battery pack, wherein the second-type of power tool battery pack has a higher nominal voltage than the first-type of power tool battery pack, and wherein the power converter is further coupled between the power input and the second battery pack interface.

[0025] In some aspects, the power converter includes an AC-to-DC converter having the first wide bandgap FET and configured to convert AC power received from the power input to DC power provided to the first battery pack interface for charging the first-type of power tool battery pack.

[0026] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency of at least 200 (kHz). In some aspects, the switching frequency is at least 400 KHz.

[0027] In some aspects, the power converter is an active clamp flyback converter.

[0028] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency between 100 kHz and 400 kHz.

[0029] AC/DC adapter assemblies described herein include a power box, a power input provided on the power box to receive AC power from an external source, an adapter coupled to the power box and including a power tool interface configured to plug into a battery pack interface of a power tool, a power converter coupled between the power input and the adapter and configured to convert power received from the power input, the power converter including a first wide bandgap field effect transistor (FET), and an electronic processor coupled to the power converter and configured control operation of the power converter by controlling the first wide bandgap FET.

[0030] In some aspects, an adapter cord couples the power box to the adapter.

[0031] In some aspects, the power tool is a heavy duty power tool configured to operate at a voltage rating between 60 V and 120 V.

[0032] In some aspects, the power converter includes an AC-to-DC converter including the first wide bandgap FET and is configured to convert AC power received from the power input to DC power provided to the adapter.

[0033] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency of at least 200 (kHz). In some aspects, the switching frequency is at least 400 kHz.

[0034] In some aspects, the power converter is an LLC converter.

[0035] In some aspects, the electronic processor controls the first wide bandgap FET at a switching frequency between 100 kHz and 400 kHz.

[0036] One embodiment provides a power converter device including a power source, a power converter coupled to the power source, and an electronic processor coupled to the power converter to control the operation of the power converter. The power converter is configured to receive an input power in one form from the power source and convert the input power to an output power in another form. The power converter includes at least one wide bandgap field effect transistors controlled by the electronic processor to convert the input power to output power.

[0037] In one example, the electronic processor controls the at least one wide bandgap field effect transistor at a switching frequency of at least 100 kHz, for example, when the power converter is an inverter. In one example, the switching frequency is at least 125 kHz.

[0038] In another example, the electronic processor controls the at least one wide bandgap field effect transistor at a switching frequency of at least 200 kHz, for example, when the power converter is a DC-to-DC converter.

[0039] In one example, the power converter is one or more selected from a group consisting of: a Full-Bridge converter, a flyback converter, an active clamp flyback converter, an LLC converter,

and a power factor correction converter.

[0040] In another example, the power converter device is one selected from a group consisting of: a power tool battery pack charger, an AC/DC adapter assembly, and a portable power supply.

[0041] Another embodiment provides a power converter device including a power source, a power converter coupled to the power source, and an electronic processor coupled to the power converter to control the operation of the power converter. The power converter configured to receive an input power in one form from the power source and convert the input power to an output power in another form. The power converter includes at least one wide bandgap FET and at least one MOSFET. The wide bandgap FET is operated at a first frequency and the MOSFET is operated at a second frequency by the electronic processor to convert the input power to output power. The first frequency is higher than the second frequency.

[0042] In one example, the power converter is a Full-Bridge converter.

[0043] In another example, the power converter device is a portable power supply.

[0044] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in its application to the details of the configuration and arrangement of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0045] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

[0046] Relative terminology, such as, for example, “about,” “approximately,” “substantially,” etc., used in connection with a quantity or condition would be understood by those of ordinary skill to be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression “from about 2 to about 4” also discloses the range “from 2 to 4”. The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%, or more) of an indicated value.

[0047] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For

example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.

[0048] Other aspects of the embodiments will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0049] FIG. 1A is a perspective view of a power converter device embodied as a power tool battery pack charger, in accordance with some embodiments.

[0050] FIG. 1B is a perspective view of a power converter device embodied as an AC/DC power adapter, in accordance with some embodiments.

[0051] FIG. 1C is a perspective view of a power converter device embodied as a portable power supply, in accordance with some embodiments.

[0052] FIG. 2 is a simplified block diagram of the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0053] FIG. 3 is a simplified block diagram of a Full-bridge converter used as a power converter in the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0054] FIG. 4 is a simplified block diagram of a Full-bridge converter used as a power converter in the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0055] FIG. 5 is a simplified block diagram of a flyback converter used as a power converter in the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0056] FIG. 6 is a simplified block diagram of an active clamp flyback converter used as a power converter in the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0057] FIG. 7 is a simplified block diagram of an LLC converter used as a power converter in the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0058] FIG. 8 is a simplified block diagram of a power factor conversion (PFC) converter used as a power converter in the power converter device of FIGS. 1A-1C, in accordance with some embodiments.

[0059] FIG. 9 is a timing diagram illustrating the difference in operation of a wide bandgap FET and a MOSFET for converting power from one form to another form, in accordance with some embodiments.

DETAILED DESCRIPTION

[0060] Power converter devices **100A-100C** are illustrated in FIGS. 1A-1C. The power converter devices **100A-100C** are singularly referred to as a power converter device **100** and collectively referred to as power converter devices **100**. The power converter devices **100** are used for various purposes associated with the operation of power tools at work sites. Further, the power converter device **100A** may be more particularly referred to as a charger **100A** (see FIG. 1A). The power converter device **100B** may be more particularly referred to as an AC/DC adapter assembly **100B** (see FIG. 1B). The power converter device **100C** may be more particularly referred to as a portable power supply **100C** (see FIG. 1C).

[0061] With reference to FIG. 1A, the charger **100A** (e.g., power converter device) is a charger for power tool battery packs and is illustrated as including a housing **105**, a first recess **110** on the

housing **105** to receive a first-type of battery pack, a second recess **115** on the housing to receive a second-type of battery pack (e.g., having a higher nominal voltage than the first-type of battery pack), and a power cord **120**. The power cord **120** can be plugged into, for example, a wall outlet to provide AC power to the charger **100A**. The charger **100A** converts the AC power to DC power and provides the DC power to charge the battery packs.

[0062] With reference to FIG. **1B**, the AC/DC adapter assembly **100B** (e.g., power converter device) is illustrated as including a power box **135**, an adapter **140**, an adapter cord **145**, and an adapter power cord **150**. The adapter power cord **150** can be plugged into, for example, a wall outlet to provide AC power to the power box **135**. The power box **135** converts the AC power to DC power and provides the DC power to operate heavy duty power tools through the adapter **140**. The heavy duty power tools are configured to operate at voltages rating from, for example, 60 V, 80 V, 120 V, and higher. The heavy duty power tools typically receive a battery pack at a battery pack interface on the power tools. The adapter **140** includes a power tool interface that mates with the battery pack interface on the power tools. The adapter cord **145** provides the DC power from the power box **135** to the adapter **140**.

[0063] With reference to FIG. **1C**, the portable power source **100C** (e.g., power converter device) is illustrated as including a housing **165**. The housing **165** has a first interface **170** provided at the top of the housing **165** to receive a first heavy duty battery pack and a second interface **175** at the bottom of the housing **165** to receive a second heavy duty battery pack. The heavy duty battery packs are, for example, battery packs used to power the heavy duty power tools as described above. The heavy duty battery packs have a nominal output voltage of, for example, at least 60 V, 80 V, 120 V, and higher. The housing **165** includes power conversion circuitry, for example, an inverter to convert the DC power from the heavy duty battery packs to AC power to be provided at AC outlets **180**. The AC outlets **180** are provided on the housing **165** to power, for example, corded AC power tools.

[0064] In some embodiments, the portable power source **100C** may include an internal non-removable battery rather than interfaces to receive battery packs. The internal battery may have a nominal voltage between 40 V and 60 V. In one embodiment, the internal battery has a nominal voltage of 50.4 V. In some embodiments the internal battery may be expandable such that battery cells may be removed or added to the internal battery by a technician. In this example, the internal battery voltage may vary between 12 V and 680 V or higher based on the configuration of the battery cells in the internal battery. In one example, the internal battery voltage may be between 12 V and 1200 V DC for a three-phase AC power source. The portable power source **100C** may include other forms of converters other than an inverter. For example, the portable power source **100C** may include a DC-DC converter as further described below. By providing a high voltage internal battery as described above, some of the electronic circuitry may be reduced. For example, providing higher voltage internal battery allows for skipping a boost DC-to-DC converter and allows for providing the voltage of the internal battery directed to the inverter. In some embodiments the portable power source **100C** illustrated in FIG. **1C** may receive lower-voltage battery packs, for example, power tool battery packs. In one example, the portable power source **100C** may receive 12 V and/or 18 V battery packs.

[0065] FIG. **2** illustrates one example embodiment of a power converter device **100** including a power source **200**, a power converter **210**, an electronic processor **220**, a memory **230**, transceiver **240**, one or more power outputs **250** (e.g., to a load), and a gate driver **260**. In some embodiments, the power source **200** is an AC power source and provides AC power to the power converter **210**. For example, the power source **200** receives AC power from the power cords **120**, **150** and provides the AC power to the power converter **210**. In other embodiments, the power source **200** is a DC power source and provides DC power to the power converter **210**. For example, the power source **200** includes the heavy duty battery packs received in the portable power source **100C** and provides the DC power from the battery packs to the power converter **210**.

[0066] The power converter **210** is configured as, for example, an inverter (shown in FIGS. **3** and **4**), a flyback converter (shown in FIG. **5**), an active clamp flyback converter (shown in FIG. **6**), an LLC converter (shown in FIG. **7**), a PFC converter (shown in FIG. **8**), and the like. The power converter **210** receives power from the power source **200** in one form (e.g., DC or AC) or at a first voltage and converts the power to another form (e.g., the other of DC or AC) or at a second voltage to provide the power at the one or more power outputs **250**. Additional details regarding the power converter **210** are provided below for each respective power converter configuration.

[0067] The memory **230** includes read only memory (ROM), random access memory (RAM), other non-transitory computer-readable media, or a combination thereof. The electronic processor **220** is configured to communicate with the memory **230** to store data and retrieve stored data. The electronic processor **220** is configured to receive instructions and data from the memory **230** and execute, among other things, the instructions. In particular, the electronic processor **220** executes instructions stored in the memory **230** to perform the functions of the power converter device **100**, including, for example, controlling switching elements of the power converter **210**. In some embodiments, the electronic processor **220** and the memory **230** are part of a microcontroller integrated circuit, rather than, for example, separate elements mounted on a circuit board. In some embodiments, the electronic processor **220** is a microprocessor with a separate memory (e.g., memory **230**). In some embodiments, the electronic processor **220** is a microcontroller including a memory and the memory **230** may not be needed or may be modified accordingly. The electronic processor **220** may be implemented as a field programmable gate array, an application specific integrated circuit, a hardware implemented state machine, and the like. In other embodiments, the electronic processor **220** may be implemented as an integrated circuit dedicated to controlling the various switching devices described herein. In some embodiments, the electronic processor **220** may be implemented using multiple processors.

[0068] The transceiver **240** allows for wired or wireless communication between the power converter device **100** and other devices, for example, the power tool battery pack, the heavy duty power tools, the heavy duty battery packs, and the like. In some embodiments, the transceiver **240** may include both a transmitter and a receiver. In other embodiments, the transceiver **240** may include a single device for transmitting and receiving.

[0069] The power converter **210** includes one or more FETs, for example, wide bandgap FETs that are driven by the gate driver **260**. The gate driver **260** receives control signals from the electronic processor **220** over a communication bus **270** to control operation of the FETs. The electronic processor **220** and the gate driver **260** provide pulse width modulated (PWM) signals to the FETs to switch the FETs at a particular frequency with a particular duty ratio depending on the conversion requirements of the power converter device **100**.

[0070] FIG. **3** illustrates one example embodiment of a Full-Bridge converter **300** that is used as the power converter **210**. In the example illustrated, the Full-Bridge converter **300** is a DC-to-AC converter that converts DC power from a power source **200** to AC power at the power output **250** (e.g., to a load). The Full-Bridge converter **300** includes a first high-side field effect transistor (FET) **330**, a first low-side FET **340**, a second high-side FET **350**, a second low-side FET **360**, an inductor **370**, and a capacitor **380**. The Full-Bridge converter **300** may include more or fewer components than those illustrated in FIG. **3**.

[0071] The drain of the first high-side FET **330** is coupled to the positive side of the power supply **200** and the source of the first high-side FET **330** is coupled to a connection point **335**. The drain of the first low-side FET **340** is coupled to the connection point **335** and the source of the first low-side FET **340** is coupled to the negative side of the power supply **200**. The drain of the second high-side FET **350** is coupled to the positive side of the power supply **200** and the source of the second high-side FET **350** is coupled to a connection point **355**. The drain of the second low-side FET **360** is coupled to the connection point **355** and the source of the second low-side FET **360** is coupled to the negative side of the power supply **200**. The gates of the FETs **330**, **340**, **350**, **360** are

coupled to the gate driver **260** controlled by the electronic processor **220**. The electronic processor **220** provides control signals to the gate driver **260** to control operation of the FETs **330**, **340**, **350**, **360**. Particularly, the electronic processor **220** controls the FETs **330**, **340**, **350**, **360** by providing a pulse width modulated (PWM) signal at the gates of the FETs **330**, **340**, **350**, **360**.

[0072] The capacitor **380** is coupled in parallel with the load **250** and the inductor **370** is coupled in series with the load **250** and the capacitor **380**. The inductor **370** is coupled on a positive side of the load **250**. The FETs **330**, **340**, **350**, **360** are controlled in a complementary fashion such that the first high-side FET **330** and the second low-side FET **360** are operated for a first half of a time period and the second high-side FET **350** and the first low-side FET **340** are operated for a second half of a time period. For the first half of the time period, the current flows from the positive side of the power source **200** to the positive side of the load **250** thereby providing a positive half of an alternating current cycle at the load **250**. For the second half of the time period, the current flows from the positive side of the power source **200** to the negative side of the load **250** thereby providing a negative half of the alternating current cycle at the load **250**.

[0073] Typically, metal oxide semiconductor field effect transistors (MOSFETs) are used in the Full-Bridge converter. MOSFETs are inefficient when operated at high frequencies and lose a high amount of energy. To dissipate the losses of the MOSFETs, additional circuitry and heat sinking features are used that add to the size and weight of the portable power supply. Also, in order to increase the efficiency, MOSFETs are operated at lower switching frequencies, as the switching frequency decreases, the size of the passive components in the circuit like inductors and capacitors will increase, which, in turn, increases the size and weight of the portable power supply. Larger heat sinks are also needed to absorb the energy lost as heat from the MOSFETs.

[0074] In the Full-Bridge converter **300**, rather than MOSFETs, wide bandgap semiconductor FETs are used for the FETs **330**, **340**, **350**, **360**. Wide bandgap semiconductors are made from, for example, Gallium Nitride (GaN), Silicon Carbide (SiC), or the like, and have a bandgaps in the range of, for example, about 3-4 electronvolts (eV). Wide bandgap semiconductors exhibit several properties that provide advantages over MOSFETs. Particularly, wide bandgap semiconductors can be operated at very high frequencies, for example, at 100 kHz, 200 kHz, 400 kHz, and more while losing less energy as heat than MOSFETs operating at lower frequencies, for example, 25 kHz, 50 kHz, and the like.

[0075] Because the wide bandgap semiconductors can be operated at very high frequencies, the power converters **210** using the wide bandgap semiconductors provide higher resolution signals at the outputs. Smaller passive components and/or filter components, that is, inductors and capacitors, can be used due to the high resolution of output signals at the power converters **210**. Additionally, smaller heat sinks and fans can be used because the wide bandgap semiconductors operate more efficiently than MOSFETs and therefore produce less heat during operation. Energy storage capacity can also be reduced for the same total output energy rating caused by the increased frequency during operation of the wide bandgap semiconductors compared to MOSFETs. Accordingly, the size and weight of the components can be reduced and efficiency of the power converter devices **100** can be improved by replacing MOSFETs with wide bandgap semiconductor devices.

[0076] FIG. **4** illustrates one example embodiment of a hybrid Full-Bridge converter **400** that can be used as the power converter **210**. In the example illustrated, the Full-Bridge converter **400** is similar in construction and operation to the Full-bridge converter **300** of FIG. **3**. The Full-Bridge **400** differs in that wide bandgap FETs are used for the first high-side FET **330** and the first low-side FET **340** and MOSFETs are used for the second high-side FET **350** and the second low-side FET **360**. In other embodiments, any combination of wide bandgap FETs and MOSFETs may be used in the Full-bridge converter **300**. The distribution of wide bandgap FETs and MOSFETs between the FETs **330**, **340**, **350**, **360** may be determined based on the desired operation of the power converter **210**.

[0077] In one example, where the low-side FETs **340, 360** are switched at a higher frequency than the high-side FETs, wide bandgap FETs are used for the low-side FETs **340, 360** and MOSFETs are used for the high-side FETs **330, 350**. Similarly, in another example in which the second FETs **350, 360** are switched at a higher frequency than the first FETs **330, 340**, wide bandgap FETs are used for the second FETs **350, 360** and MOSFETs are used for the first FETs **330, 340**. In other words, where some FETs in a converter are switched at a higher frequency than other FETs, the FETs switched at a higher frequency may be replaced with the wide bandgap FETs because, as noted earlier, wide bandgap switches are configured to switch at a higher rate with less accommodations (e.g., for space and heat generation) than MOSFETs. Accordingly, in some embodiments, the desired operation of the power converter **210**, heat generation reduction, and size reduction, may still be fulfilled by using one or more wide bandgap FETs together with one or more MOSFETs in the power converter **210**, while reducing the cost of the power converter **210** relative to a power converter **210** that uses all wide bandgap FETs, which are typically more expensive than MOSFETs.

[0078] FIG. **5** illustrates one example embodiment of a flyback converter **500** that can be used in the power converter **210**. The flyback converter **500** may be used as a boost or buck DC-to-DC converter within the power converter **210**. In the example illustrated, the flyback converter **500** receives power from the power source **200** and includes a primary side circuit **510** and a secondary side circuit **520** coupled by a two winding inductor **530**. The flyback converter **500** may include more or fewer components than those illustrated in FIG. **5**.

[0079] The primary side circuit **510** includes a flyback switching FET **540** coupled in series with a primary side of the two winding inductor **530** and an input capacitor **550** coupled in parallel to the flyback switching FET **540** and the two winding inductor **530**. The input capacitor **550** is coupled in parallel to the power source **200**. A passive clamp resistor-capacitor-diode (RCD) snubber circuit **580** may also be provided in series with the flyback switching FET **540** and in parallel to the primary side of the two winding inductor **530**. The passive clamp RCD snubber circuit limits voltage spikes on the flyback switching FET **540**. The secondary side circuit **520** includes an output diode **560** and an output capacitor **570**. The output diode **560** is coupled in series on a positive side of a secondary side of the two winding inductor **530**. The output capacitor **570** is coupled in parallel to the secondary side of the two winding inductor **530** downstream of the output diode **560**. A load **250** is coupled in parallel to the output capacitor **570**.

[0080] During operation, the flyback switching FET **540** is turned on to store energy from the power source **200** on the two winding inductor **530**. During the on-time of the flyback switching FET **540**, the output capacitor **570** provides the operating power to the load **250**. When the flyback switching FET **540** is turned off, the energy stored on the two winding inductor **530** is transferred to the output capacitor **570** and the load **250**. In the example illustrated, a wide bandgap FET is used as the flyback switching FET **540** to enable high-frequency operation. The flyback switching FET **540** is driven by a gate driver **260** controlled by the electronic processor **220**. The electronic processor **220** provides control signals to the gate driver **260** to control the operation of the flyback switching FET **540**. In some embodiments, a wide bandgap FET may be used to replace the diode **560** to further increase the frequency of operation for the flyback converter **500**.

[0081] FIG. **6** illustrates one example embodiment of an active clamp flyback converter **600** that can be used in the power converter **210**. The active clamp flyback converter **600** may be used as a boost or buck DC-to-DC converter within the power converter **210**. In the example illustrated, the active clamp flyback converter **600** receives power from the power source **200** and includes a primary side circuit **610** and a secondary side circuit **620** coupled by a two winding inductor **630**. The active clamp flyback converter **600** may include more or fewer components than those illustrated in FIG. **6**.

[0082] The primary side circuit **610** includes a flyback switching FET **640** coupled in series with a primary side of the two winding inductor **630** and an input capacitor **650** coupled in parallel to the

flyback switching FET **640** and the two winding inductor **630**. The input capacitor **650** is coupled in parallel to the power source **200**. The secondary side circuit **620** includes an output diode **660** and an output capacitor **670**. The output diode **660** is coupled in series on a positive side of a secondary side of the two winding inductor **630**. The output capacitor **670** is coupled in parallel to the secondary side of the two winding inductor **630** downstream of the output diode **660**. A load **250** is coupled in parallel to the output capacitor **670**.

[0083] The primary side circuit **610** also includes an active clamp circuit **690** including a clamp FET **694** and a clamp capacitor **698** connected in series. The active clamp circuit **690** is connected in parallel with the primary side of the two winding inductor **630** and in series with the flyback switching FET **640**.

[0084] During operation, the flyback switching FET **640** is turned on to store energy from the power source **200** on the two winding inductor **630**. During the on-time of the flyback switching FET **640**, the output capacitor **670** provides the operating power to the load **250**. When the flyback switching FET **640** is turned off, the energy stored on the two winding inductor **630** is transferred to the output capacitor **670** and the load **250**. The clamp FET **694** is turned on during the off state of the flyback switching FET **640** to reduce the stress on the flyback switching FET **640**. In the example illustrated, a wide bandgap FET is used for one or more of the flyback switching FET **640**, the output diode **660**, and the clamp FET **694** based on the desired characteristics of the power converter **210**. Particularly, by using wide bandgap FETs, the size of the inductive and capacitor components may be reduced compared to an active clamp flyback converter using MOSFETs for the flyback switching FET **640** and the clamp FET **694**. The flyback switching FET **640** and the clamp FET **694** are driven by a gate driver **260** controlled by the electronic processor **220**. The electronic processor **220** provides control signals to the gate driver **260** to control the operation of the flyback switching FET **640** and the clamp FET **694**.

[0085] FIG. 7 illustrates one example embodiment of an LLC converter **700** that can be used as the power converter **210**. In the example illustrated, the LLC converter includes a transformer **705** connecting a primary side circuit **710** to a secondary side circuit **715**. The primary side circuit **710** includes a half bridge **720** and a resonant tank **725**. The secondary side circuit **715** includes a first output diode **730**, a second output diode **735**, and an output capacitor **740**. The transformer **705** includes one primary winding **745** connected to the primary side circuit **710** and a first secondary windings **750** and second secondary winding **755** connected to the secondary side circuit **715**. The LLC converter **700** may include more or fewer components than those illustrated in FIG. 7.

[0086] The half bridge **720** includes a high-side FET **760** and a low side FET **765** connected in series with the source of the high-side FET **760** connected to a drain of the low side FET **765** at a connection point **770**. The FETs **760**, **765** are driven by a gate driver **260** controlled by the electronic processor **220**. The electronic processor **220** provides control signals to the gate driver **260** to operate the FETs **760**, **765**. The half bridge **720** is coupled in parallel to the power source **200**. In some embodiments, the primary side circuit **710** may include a full bridge with additional high-side and low-side FETs.

[0087] The resonant tank **725** includes a capacitor **775** and a first inductor **780** connected in series and to the connection point **770**. The capacitor **775** and the first inductor **780** are also in series with the primary winding **745**. The resonant tank **725** also includes a second inductor **785**, which is a magnetizing inductance of the transformer **705**, and is connected in series to the capacitor **775** and the first inductor **780** and in parallel to the primary winding **745**.

[0088] The first output diode **730** is connected in series with the first secondary winding **750** and the second output diode **735** is connected in series with the second secondary winding **755**. The output capacitor **740** is connected in parallel to the first secondary winding **750** and the second secondary winding **755**. The output capacitor **740** is also connected in parallel to a load **250**. In some embodiments, the secondary side circuit **715** may also include a full bridge of diodes.

[0089] During operation, the electronic processor **220** controls the switching of the high-side FET

760 and the low-side FET **765** to convert AC voltage from the power source **200** to DC voltage at the load **250**. Wide bandgap FETs are used instead of MOSFETs for the high-side FET **760** and the low-side FET **765**. Use of the wide bandgap FETs results in an LLC converter that is smaller in size and more efficient than an LLC converter including MOSFETs. Wide bandgap devices may also be used for the diodes in the secondary side circuit **715**.

[0090] FIG. **8** illustrates one example embodiment of a power factor correction (PFC) converter **800** that can be used as the power converter **210**. The PFC converter **800** includes a diode rectifier **810**, an inductor **820**, a converter FET **830**, an output diode **840**, an output capacitor **850**, and a gate driver **260**. The diode rectifier **810** receives AC input voltage from an AC power source, for example, the power source **200**. The diode rectifier **810** rectifies the AC voltage into a rectified AC voltage at the output.

[0091] The inductor **820** is connected in series with the diode rectifier **810** to receive the rectified voltage. The output diode **840** and the output capacitor **850** are connected in series with the inductor **820**. The load **250** is connected in parallel to the output capacitor **850** and in series with the output diode **840**. The converter FET **830** is connected in series with the inductor **820** and in parallel to the output diode **840** and the output capacitor **850**.

[0092] The gate driver **260** drives the converter FET **830** to operate the PFC converter **800**. The gate driver **260** receives control signals from the electronic processor **220** to control the operation of the converter FET **830**. The electronic processor **220** implements a power factor control algorithm to drive the converter FET **830** and convert AC voltage to DC voltage at the load **250**. A wide bandgap FET is used for the converter FET **830** to increase efficiency and reduce the size of the PFC converter **800**.

[0093] FIG. **9** illustrates a timing chart **900** of one example implementation of the power converter **210** showing the differences between the operation of a MOSFET and a wide bandgap FET in the power converter **210**. The power converter **210** is used in this implementation to convert DC power to an AC output at 60 Hz. The output AC signal **910** is illustrated at the bottom of the timing chart **900** and includes a sine wave output having a frequency of 60 Hz. In a power converter **210** using a MOSFET, the MOSFET is controlled at a frequency (e.g., a second frequency) similar to the output frequency of the AC output. That is, the MOSFET is controlled at 60 Hz to convert the DC input to the AC output. A PWM signal of 60 Hz is input to the gate driver to drive the MOSFET. The MOSFET timing waveform **920** is shown at the top of the timing chart **900**.

[0094] In a power converter **210** using a wide bandgap FET, the wide bandgap FET is controlled at a frequency (e.g., a first frequency) much higher than the output frequency of the AC output of the frequency of the PWM signal provided to the MOSFETs. In one example, the wide bandgap FETs may be controlled by providing a PWM signal having a frequency between 100 kHz-400 kHz. The wide bandgap FET timing waveform **930** is shown in the middle of the timing chart **900**. A zoomed in version during a positive phase **940** illustrates the respective signals during a positive phase of the AC output signal. A zoomed in version during a negative phase **950** illustrates the respective signals during a negative phase of the AC output signal. The duty ratio of the PWM signal provided to the wide bandgap FET during the positive phase (e.g., 90%) is much higher than the duty ratio during the negative phase (e.g., 10%). Controlling the FETs with high frequency allows for a smoother AC output signal and improves the efficiency of the power converter **210**.

[0095] Thus, various embodiments described herein provide for power converter devices having wide bandgap semiconductors. Various features and advantages are set forth in the following claims.

Claims

1. A power supply system comprising: a housing; an internal non-removable battery received in the housing and having a nominal voltage of at least 12 Volts; an output provided on the housing; a

- power converter coupled between the internal non-removable battery and the output and configured to convert power received from the internal non-removable battery, the power converter including a first switch; and an electronic processor coupled to the power converter and configured to control operation of the power converter by controlling the first switch at a switching frequency of at least 100 kHz.
2. The power supply system of claim 1, wherein the power converter includes an inverter including the first switch and is configured to convert direct-current (DC) power received from the internal non-removable battery to alternating-current (AC) power, wherein the AC power is provided at the output.
 3. The power supply system of claim 2, wherein the power converter further includes a DC-to-DC converter configured to: convert the DC power received from the internal non-removable battery at a first voltage to DC power at a second voltage, and provide the DC power at the second voltage to the inverter, wherein the DC-to-DC converter includes a second switch.
 4. The power supply system of claim 3, wherein the electronic processor controls the second switch at a switching frequency of at least 200 kHz.
 5. The power supply system of claim 2, wherein the inverter includes: a first bridge circuit having a first high-side FET and a first low-side FET; and a second bridge circuit having a second high-side FET and a second low-side FET, wherein at least one of the first high-side FET, the first low-side FET, the second high-side FET, and the second low-side FET is the first switch, and wherein at least another of the first high-side FET, the first low-side FET, the second high-side FET, and the second low-side FET is a metal oxide semiconductor field effect transistor (MOSFET).
 6. The power supply system of claim 5, wherein: the first high-side FET is the first switch and the first low-side FET is a second switch; and the second high-side FET and the second low-side FET are MOSFETs.
 7. The power supply system of claim 5, wherein: the first high-side FET is the first switch and the second high-side FET is a second switch; and wherein the first low-side FET and the second low-side FET are MOSFETs.
 8. The power supply system of claim 5, wherein: the first low-side FET is the first switch and the second low-side FET is a second switch; and wherein the first high-side FET and the second high-side FET are MOSFETs.
 9. The power supply system of claim 5, wherein: the electronic processor operates the first switch at a first frequency and operates the MOSFET at a second frequency; and wherein the first frequency is higher than the second frequency.
 10. The power supply system of claim 1, wherein: the power converter includes a DC-to-DC converter having the first switch and is configured to convert DC power received from the internal non-removable battery at a first voltage to a second voltage; and the DC power at the second voltage is provided at the output.
 11. The power supply system of claim 1, wherein the electronic processor controls the first switch at a switching frequency between 100 kHz and 400 kHz.
 12. The power supply system of claim 1, wherein the internal non-removable battery has a nominal voltage of between 12 Volts and 680 Volts.
 13. A power supply system comprising: a battery power source having a nominal voltage of at least 12 V; an output; an inverter coupled between the battery power source and the output and configured to convert power received from the battery power source, the inverter including a first wide bandgap field effect transistor (FET); and an electronic processor coupled to the inverter and configured to control operation of the inverter by controlling the first wide bandgap FET at a switching frequency of at least 100 kHz.
 14. The power supply system of claim 13, wherein the electronic processor controls the first wide bandgap FET at a switching frequency of at least 200 kHz.
 15. The power supply system of claim 13, wherein the inverter includes: a first bridge circuit

having a first high-side FET and a first low-side FET; and a second bridge circuit having a second high-side FET and a second low-side FET, wherein at least one of the first high-side FET, the first low-side FET, the second high-side FET, and the second low-side FET is the first wide bandgap FET, and wherein at least another of the first high-side FET, the first low-side FET, the second high-side FET, and the second low-side FET is a metal oxide semiconductor field effect transistor (MOSFET).

16. The power supply system of claim 15, wherein: the electronic processor operates the first wide bandgap FET at a first frequency and operates the MOSFET at a second frequency; and wherein the first frequency is higher than the second frequency.

17. The power supply system of claim 13, wherein the electronic processor controls the first wide bandgap FET at a switching frequency between 100 kHz and 400 kHz.

18. The power supply system of claim 13, wherein the battery power source has a nominal voltage of between 12 Volts and 680 Volts.

19. A power supply system comprising: a battery power source; an input configured to receive AC power from an external source; an output; an inverter coupled between the battery power source and the output and configured to convert power received from the battery power source, the inverter including a first wide bandgap field effect transistor (FET); a power converter coupled between the input and the battery power source and configured to convert the AC power received from the input, the power converter including a second wide bandgap FET; and an electronic processor coupled to the inverter and configured to control operation of the inverter by controlling the first wide bandgap FET at a switching frequency of at least 100 kHz.

20. The power supply system of claim 19, wherein the power converter includes an AC-to-DC converter having the second wide bandgap FET and configured to convert the AC power received from the input to DC power provided to the battery power source for charging the power source.
